

The graph is acyclic, thus the most efficient algorithm is SP-Tree Acyclic whose complexity is $O(m)$.

Preprocessing: all the nodes satisfy the topological order
there is no need to renumber them

$$d[1] = 0$$

$$d[2] = 0 + 1$$

$$d[3] = \min\{0 + 0, 1 - 4\} = -3$$

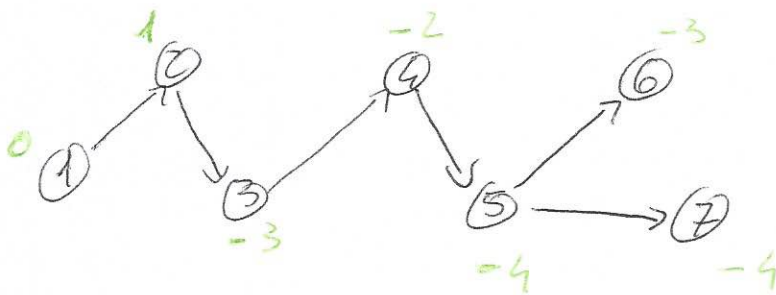
$$d[4] = \min\{1 + 1, -3 + 1\} = -2$$

$$d[5] = \min\{-3 + 0, -2 - 2\} = -4$$

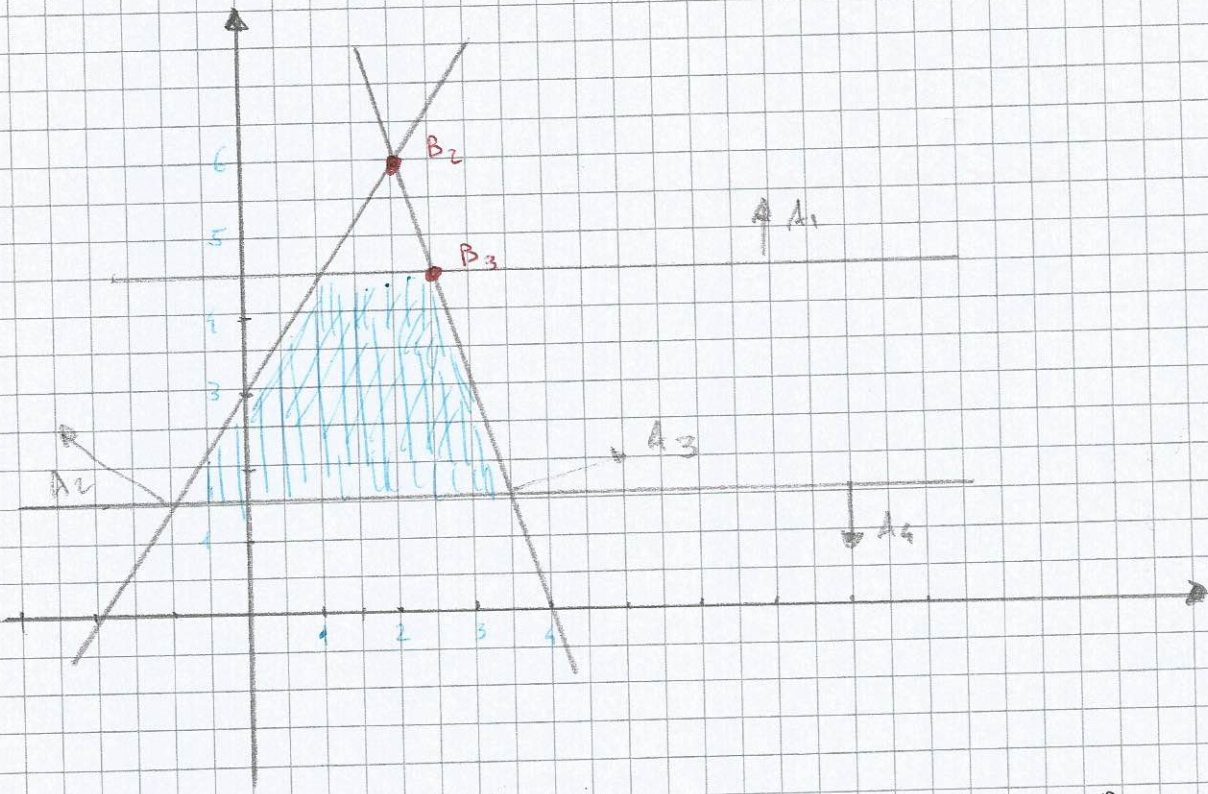
$$d[6] = \min\{-2 + 2, -4 + 1\} = -3$$

$$d[7] = \min\{-3 - 1, -4 + 0\} = -4$$

Final shortest path tree



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$B_1 = \{1, 4\}$ $A_{B1} = \begin{bmatrix} 0 & 2 \\ 0 & -4 \end{bmatrix}$ has rank < 2 B_1 is not a base

$B_2 = \{2, 3\}$ $A_{B2} = \begin{bmatrix} -3 & 2 \\ 3 & 1 \end{bmatrix}$ $A_{B2}^{-1} = \begin{bmatrix} -1/9 & 2/9 \\ 1/3 & 1/3 \end{bmatrix}$
 $A_N = \begin{bmatrix} 0 & 2 \\ 0 & -4 \end{bmatrix}$ $b_B = \begin{bmatrix} 6 \\ 12 \end{bmatrix}$ $b_N = \begin{bmatrix} 9 \\ 9 \end{bmatrix}$

Primal feasibility check:

$$A_N A_B^{-1} b_B \leq b_N \quad \begin{bmatrix} 0 & 2 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} -1/9 & 2/9 \\ 1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 12 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 12 \\ -24 \end{bmatrix} \leq \begin{bmatrix} 9 \\ 9 \end{bmatrix}$$

the first constraint is violated

Dual feasibility check

$$c A_B^{-1} \geq 0 \quad \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} -1/9 & 2/9 \\ 1/3 & 1/3 \end{bmatrix} = \begin{bmatrix} \frac{4}{9} & \frac{7}{9} \end{bmatrix} \geq 0$$

dual feasible

$$B_3 = \{1, 3\}$$

$$A_{B_3} = \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix}$$

$$A_{B_3}^{-1} = \begin{bmatrix} -1/6 & 1/3 \\ 1/2 & 0 \end{bmatrix}$$

$$A_N = \begin{bmatrix} -3 & 2 \\ 0 & -4 \end{bmatrix}$$

$$b_B = \begin{bmatrix} 9 \\ 12 \end{bmatrix}$$

$$b_N = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

Primal feasibility check

$$A_N A_B^{-1} b_B \leq b_N$$

$$\begin{bmatrix} -3 & 2 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} -1/6 & 1/3 \\ 1/2 & 0 \end{bmatrix} \begin{bmatrix} 9 \\ 12 \end{bmatrix} \leq \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 2 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} 5/2 \\ 3/2 \end{bmatrix} = \begin{bmatrix} 3/2 \\ -18 \end{bmatrix} \leq \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

primal feasible

Dual feasibility check

$$c A_B^{-1} = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} -1/6 & 1/3 \\ 1/2 & 0 \end{bmatrix} = \begin{bmatrix} 1/6 & 2/3 \end{bmatrix} \geq 0$$

dual feasible

B_3 is an optimal base

$$\text{Max } 35x_1 + 21x_2 + 11x_3 + 9x_4 + 3x_5 - 6x_6$$

$$7x_1 + 5x_2 + 4x_3 + 2x_4 - 3x_5 + x_6 \leq 8$$

$$x_i \in \{0, 1\} \quad i=1..6$$

Q3

Fixing $x_5 = 1$ since the weight of 5 is < 0 and the value is > 0

$x_6 = 0$ since the weight of 6 is > 0 and the ~~value is~~ > 0 value is < 0

The problem reduces to

$$\begin{aligned} P_0: \text{max } & 35x_1 + 21x_2 + 11x_3 + 9x_4 \\ & 7x_1 + 5x_2 + 4x_3 + 2x_4 \leq 11 \\ & x_i \in \{0, 1\} \quad i=1..4 \end{aligned}$$

Ratio $\frac{\text{value}}{\text{weight}}$

$$\frac{35}{7} = 5 \quad \frac{21}{5} = 4.2 \quad \frac{11}{4} = 2.75 \quad \frac{9}{2} = 4.5$$

order 1 3 4 2

ub of P_0 capacity

$$x_1 = 1 \quad 11 \rightarrow 4$$

$$x_4 = 1 \quad 4 \rightarrow 2$$

$$x_2 = 2/5 \quad 2 \rightarrow 0$$

$$x_3 = 0$$

$$\text{value} = 35 + 9 + \frac{42}{5} = 52 + 21.5 \rightarrow 52$$

lb

$$x_1 = 1$$

$$x_4 = 1$$

$$x_2 = 0$$

$$x_3 = 0$$

$$\text{value } 35 + 9 = 44 \quad \text{incumbent}$$

$$\begin{aligned} P_1: & 21 + \text{max } 35x_1 + 11x_3 + 9x_4 \\ & 7x_1 + 4x_3 + 2x_4 \leq 6 \end{aligned}$$

Fixing BY FEASIBILITY $x_1 = 0$

ub of P_1

$$x_4 = 1$$

$$x_3 = 1$$

feasible

$$\text{value} = 21 + 9 + 11 = 41$$

$$P_2: \max 35x_1 + 11x_3 + 9x_4$$

$$7x_1 + 4x_3 + 2x_4 \leq 11$$

$$\text{ub } \begin{array}{ll} x_1 = 1 & 11 \rightarrow 4 \\ x_4 = 1 & 1 \rightarrow 2 \\ x_3 = 2/4 & 2 \rightarrow 0 \end{array}$$

$$\text{value } 35 + 9 + \frac{11}{2} = 49 + 1/2 \rightarrow 49$$

$$P_3: 11 + \max 35x_1 + 9x_4$$

$$7x_1 + 2x_4 \leq 7$$

$$\text{ub: } \begin{array}{ll} x_1 = 1 & \text{feasible} \\ x_4 = 0 & \text{ub=lb} \end{array}$$

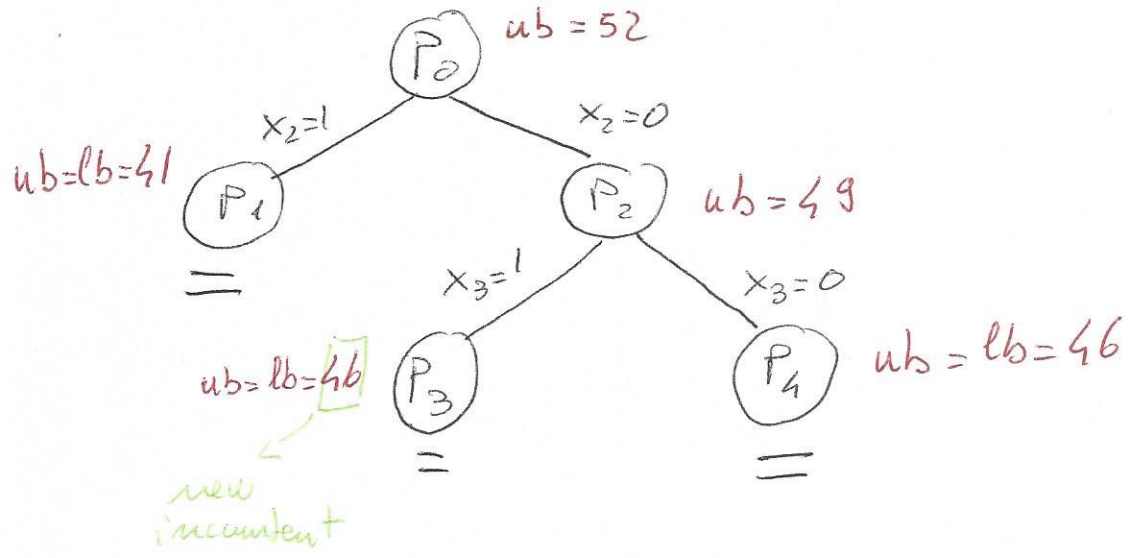
$$\text{value } 11 + 35 = 46 \text{ new incumbent}$$

$$P_4: \max 35x_1 + 9x_4$$

$$7x_1 + 2x_4 \leq 11$$

$$\text{ub: } \begin{array}{ll} x_1 = 1 & \text{feasible} \\ x_4 = 1 & \text{ub=lb} \end{array}$$

$$\text{value} = 35 + 9 = 44$$



- i) false . The feasible region of $\bar{P}$ includes all the feasible points of P and other points with fractional coordinates satisfying the constraints.
- ii) true . If the optimal solution of $\bar{P}$ has all integer components it satisfies also all the constraints of P . Since the objective function is the same in both problems, the value is the same.
- iii) false . Rounding the components of the optimal solution of $\bar{P}$ does not guarantee that the solution satisfies all the constraints.

Set I ;

Set J ;

param $b \{I, J\}$;

var $x \{I, J\}$ integer;

maximize obj:

$\text{sum } \{i \text{ in } I, j \text{ in } J : b[i, j] > 0\} x[i, j];$

x_{ij} = flow on arc $(i,j) \in A$ ≥ 0

$$y_{ij} = \begin{cases} 1 & \text{if } x_{ij} > 0 \\ 0 & \text{if } x_{ij} = 0 \end{cases}$$

Flow conservation constraints

$$\sum_{(j,i) \in BS(i)} x_{ji} - \sum_{(i,j) \in FS(i)} x_{ij} = \begin{cases} -s_i & \text{if } i \in S \\ z_i & \text{if } i \in C \\ 0 & \text{if } i \in T \end{cases} \quad \forall i \in N$$

Capacity / Linking constraints

$$x_{ij} \leq u_{ij} y_{ij} \quad \forall (i,j) \in A$$

objective function

$$\min \sum_{(i,j) \in A} c_{ij} y_{ij}$$

Variant

$$\min \sum_{(i,j) \in A} c_{ij} y_{ij} + \sum_{i \in T} p_i \left(\sum_{(i,j) \in FS(i)} x_{ij} \right)$$

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min d

$$d_j \quad d - \sum_{i=1}^n c_{ij} x_{ij} \geq 0 \quad j=1 \dots m$$

$$\beta_i \quad - \sum_{j=1}^m a_{ij} x_{ij} - b_i y_i \leq 0 \quad i=1 \dots n$$

$$\gamma \quad \sum_{i=1}^n y_i = 1$$

$$d \geq 0 \quad y_i \geq 0 \quad i=1 \dots n$$

max γ

$$\sum_{j=1}^m d_j \leq 1$$

$$-c_{ij} d_j + a_{ij} \beta_i = 0 \quad i=1 \dots n \quad j=1 \dots m$$

$$-b_i \beta_i + \gamma \leq 0 \quad i=1 \dots n$$

$$d_j \geq 0 \quad j=1 \dots m$$

$$\beta_i \leq 0 \quad i=1 \dots n$$

$$\gamma \geq 0$$

CS conditions

$$d_j \left(d - \sum_{i=1}^n c_{ij} x_{ij} \right) = 0 \quad j=1 \dots m$$

$$\beta_i \left(\sum_{j=1}^m a_{ij} x_{ij} - b_i y_i \right) = 0 \quad i=1 \dots n$$

$$d \left(\sum_{j=1}^m d_j - 1 \right) = 0$$

$$y_i \left(-b_i \beta_i - \gamma \right) = 0 \quad i=1 \dots n$$